# Results

## Diagnostic Accuracy Calculator (Python)

Diagnostic accuracy from entered independent-observation counts. Sample predictive values and any supplied-prevalence predictive values are reported separately.

Confusion Counts

| Classification | Count |
|---|---|
| True positive | 90 |
| False positive | 30 |
| False negative | 20 |
| True negative | 80 |

Sample Accuracy

| Metric | Estimate | CI Lower | CI Upper |
|---|---|---|---|
| Sensitivity | 81.8% | 73.3% | 88.5% |
| Specificity | 72.7% | 63.4% | 80.8% |
| Positive predictive value | 75.0% | 66.3% | 82.5% |
| Negative predictive value | 80.0% | 70.8% | 87.3% |
| Accuracy | 77.3% | 71.2% | 82.6% |

*Note.* 95% exact Clopper-Pearson intervals assume independent observations and a fixed classification rule. PPV, NPV and accuracy describe this sample, not population performance in a case-control sample.

Diagnostic Ratios and Youden Index

| Metric | Estimate | Value or Status |
|---|---|---|
| Positive likelihood ratio | 3.000 | 3 |
| Negative likelihood ratio | 0.250 | 0.25 |
| Diagnostic odds ratio | 12.000 | 12 |
| Youden index | 0.545 | 0.545455 |

*Note.* No continuity correction is applied. Infinite means a positive numerator with a zero denominator; undefined includes zero divided by zero or a missing reference margin. Ratio intervals are omitted.

## Sequential Testing (Python)

> Conditional independence is an assumption, not estimated from data. Use pycotest when modelling conditional dependence.

Both tests are assumed conditionally independent within each reference group. Serial positive requires both tests positive. Serial negative retests first-test negatives and uses either-positive classification; parallel applies both tests to everyone. Entered test properties are treated as known; no uncertainty intervals are propagated. Costs include testing only, in the entered cost units.

Combined Accuracy

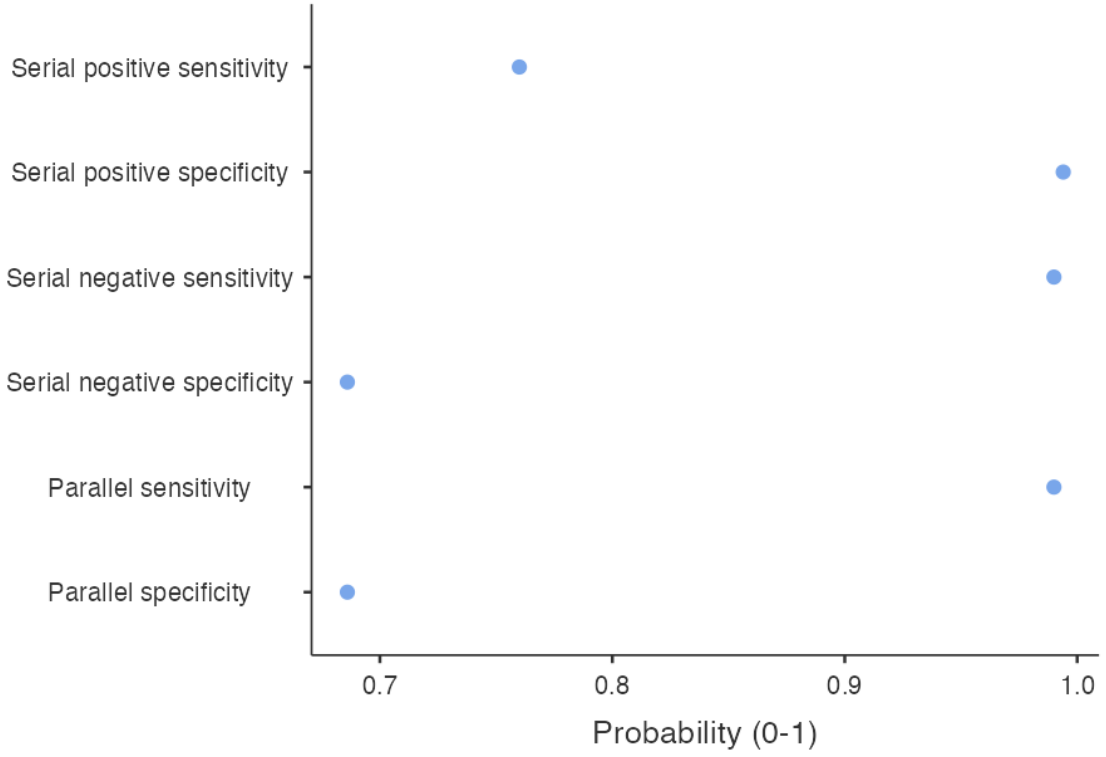
| Strategy | Sensitivity | Specificity | PPV | NPV | LR+ | LR- |
|---|---|---|---|---|---|---|
| Serial positive | 76.0% | 99.4% | 93.4% | 97.4% | 126.667 | 0.241449 |
| Serial negative | 99.0% | 68.6% | 25.9% | 99.8% | 3.15287 | 0.0145773 |
| Parallel | 99.0% | 68.6% | 25.9% | 99.8% | 3.15287 | 0.0145773 |

Expected Counts and Workload

| Strategy | TP | FP | FN | TN | First Tests | Second Tests |
|---|---|---|---|---|---|---|
| Serial positive | 76.0 | 5.40 | 24.00 | 895 | 1000 | 365 |
| Serial negative | 99.0 | 282.60 | 1.00 | 617 | 1000 | 635 |
| Parallel | 99.0 | 282.60 | 1.00 | 617 | 1000 | 1000 |

*Note.* Expected counts are fractional projections, not observed participants. Serial negative and parallel have identical accuracy under the model, but different second-test workload.

**Combined Sensitivity and Specificity**




## Benford Analysis (Python)

> Expected frequencies below 5 make the chi-square approximation potentially unreliable. Consider fewer leading digits or a larger suitable sample.

Positive finite measurements: 36. Excluded missing: 0; zero: 0; negative: 0. Use only naturally occurring measurements for which a leading-digit model is defensible; identifiers, imposed ranges and detection limits can invalidate this comparison. Mantissa arc tests and suspect-record classifications are outside this core.

Leading-Digit Goodness of Fit

| N | Digits | Pearson Chi-Square | df | Approximate p | Mean Absolute Deviation | Orders of Magnitude |
|---|---|---|---|---|---|---|
| 36 | 2 | 340 | 89 | <.001 | 0.0191 | 5.00 |

*Note.* MAD is descriptive; no fixed conformity label is applied. The p-value is asymptotic. Departure from Benford frequencies does not establish fraud or identify its cause.

Leading-Digit Frequencies

| Leading Digits | Observed | Expected Count | Observed Proportion | Expected Proportion |
|---|---|---|---|---|
| 10 | 6 | 1.490 | 16.7% | 4.1% |
| 11 | 0 | 1.360 | 0.0% | 3.8% |
| 12 | 0 | 1.251 | 0.0% | 3.5% |
| 13 | 5 | 1.159 | 13.9% | 3.2% |
| 14 | 0 | 1.079 | 0.0% | 3.0% |
| 15 | 0 | 1.009 | 0.0% | 2.8% |
| 16 | 0 | 0.948 | 0.0% | 2.6% |
| 17 | 0 | 0.894 | 0.0% | 2.5% |
| 18 | 0 | 0.845 | 0.0% | 2.3% |
| 19 | 5 | 0.802 | 13.9% | 2.2% |
| 20 | 0 | 0.763 | 0.0% | 2.1% |
| 21 | 0 | 0.727 | 0.0% | 2.0% |
| 22 | 0 | 0.695 | 0.0% | 1.9% |
| 23 | 0 | 0.665 | 0.0% | 1.8% |
| 24 | 0 | 0.638 | 0.0% | 1.8% |
| 25 | 0 | 0.613 | 0.0% | 1.7% |
| 26 | 5 | 0.590 | 13.9% | 1.6% |
| 27 | 0 | 0.569 | 0.0% | 1.6% |
| 28 | 0 | 0.549 | 0.0% | 1.5% |
| 29 | 0 | 0.530 | 0.0% | 1.5% |
| 30 | 0 | 0.513 | 0.0% | 1.4% |
| 31 | 0 | 0.496 | 0.0% | 1.4% |
| 32 | 0 | 0.481 | 0.0% | 1.3% |
| 33 | 0 | 0.467 | 0.0% | 1.3% |
| 34 | 0 | 0.453 | 0.0% | 1.3% |
| 35 | 0 | 0.440 | 0.0% | 1.2% |
| 36 | 0 | 0.428 | 0.0% | 1.2% |
| 37 | 5 | 0.417 | 13.9% | 1.2% |
| 38 | 0 | 0.406 | 0.0% | 1.1% |
| 39 | 0 | 0.396 | 0.0% | 1.1% |
| 40 | 0 | 0.386 | 0.0% | 1.1% |
| 41 | 0 | 0.377 | 0.0% | 1.0% |
| 42 | 0 | 0.368 | 0.0% | 1.0% |
| 43 | 0 | 0.359 | 0.0% | 1.0% |
| 44 | 0 | 0.351 | 0.0% | 1.0% |
| 45 | 0 | 0.344 | 0.0% | 1.0% |
| 46 | 0 | 0.336 | 0.0% | 0.9% |
| 47 | 0 | 0.329 | 0.0% | 0.9% |
| 48 | 0 | 0.322 | 0.0% | 0.9% |
| 49 | 0 | 0.316 | 0.0% | 0.9% |
| 50 | 0 | 0.310 | 0.0% | 0.9% |
| 51 | 5 | 0.304 | 13.9% | 0.8% |

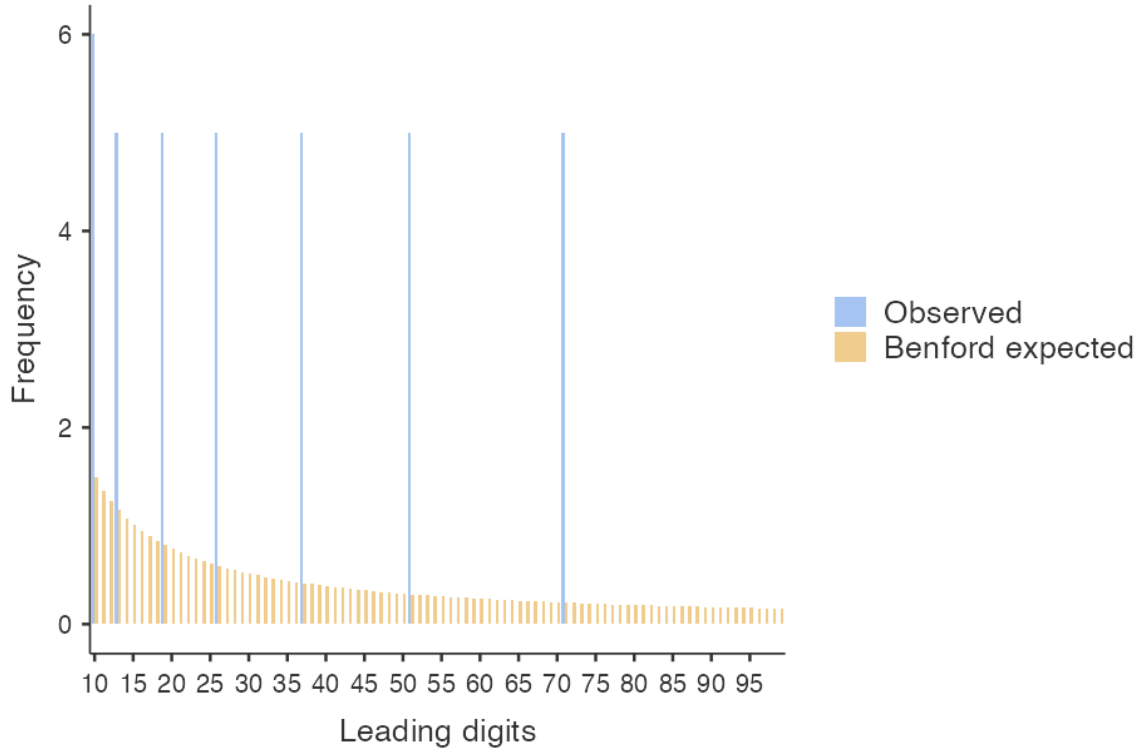
Leading-Digit Frequencies

| Leading Digits | Observed | Expected Count | Observed Proportion | Expected Proportion |
|---|---|---|---|---|
| 52 | 0 | 0.298 | 0.0% | 0.8% |
| 53 | 0 | 0.292 | 0.0% | 0.8% |
| 54 | 0 | 0.287 | 0.0% | 0.8% |
| 55 | 0 | 0.282 | 0.0% | 0.8% |
| 56 | 0 | 0.277 | 0.0% | 0.8% |
| 57 | 0 | 0.272 | 0.0% | 0.8% |
| 58 | 0 | 0.267 | 0.0% | 0.7% |
| 59 | 0 | 0.263 | 0.0% | 0.7% |
| 60 | 0 | 0.258 | 0.0% | 0.7% |
| 61 | 0 | 0.254 | 0.0% | 0.7% |
| 62 | 0 | 0.250 | 0.0% | 0.7% |
| 63 | 0 | 0.246 | 0.0% | 0.7% |
| 64 | 0 | 0.242 | 0.0% | 0.7% |
| 65 | 0 | 0.239 | 0.0% | 0.7% |
| 66 | 0 | 0.235 | 0.0% | 0.7% |
| 67 | 0 | 0.232 | 0.0% | 0.6% |
| 68 | 0 | 0.228 | 0.0% | 0.6% |
| 69 | 0 | 0.225 | 0.0% | 0.6% |
| 70 | 0 | 0.222 | 0.0% | 0.6% |
| 71 | 5 | 0.219 | 13.9% | 0.6% |
| 72 | 0 | 0.216 | 0.0% | 0.6% |
| 73 | 0 | 0.213 | 0.0% | 0.6% |
| 74 | 0 | 0.210 | 0.0% | 0.6% |
| 75 | 0 | 0.207 | 0.0% | 0.6% |
| 76 | 0 | 0.204 | 0.0% | 0.6% |
| 77 | 0 | 0.202 | 0.0% | 0.6% |
| 78 | 0 | 0.199 | 0.0% | 0.6% |
| 79 | 0 | 0.197 | 0.0% | 0.5% |
| 80 | 0 | 0.194 | 0.0% | 0.5% |
| 81 | 0 | 0.192 | 0.0% | 0.5% |
| 82 | 0 | 0.190 | 0.0% | 0.5% |
| 83 | 0 | 0.187 | 0.0% | 0.5% |
| 84 | 0 | 0.185 | 0.0% | 0.5% |
| 85 | 0 | 0.183 | 0.0% | 0.5% |
| 86 | 0 | 0.181 | 0.0% | 0.5% |
| 87 | 0 | 0.179 | 0.0% | 0.5% |
| 88 | 0 | 0.177 | 0.0% | 0.5% |
| 89 | 0 | 0.175 | 0.0% | 0.5% |
| 90 | 0 | 0.173 | 0.0% | 0.5% |
| 91 | 0 | 0.171 | 0.0% | 0.5% |
| 92 | 0 | 0.169 | 0.0% | 0.5% |
| 93 | 0 | 0.167 | 0.0% | 0.5% |

Leading-Digit Frequencies

| Leading Digits | Observed | Expected Count | Observed Proportion | Expected Proportion |
|---|---|---|---|---|
| 94 | 0 | 0.165 | 0.0% | 0.5% |
| 95 | 0 | 0.164 | 0.0% | 0.5% |
| 96 | 0 | 0.162 | 0.0% | 0.5% |
| 97 | 0 | 0.160 | 0.0% | 0.4% |
| 98 | 0 | 0.159 | 0.0% | 0.4% |
| 99 | 0 | 0.157 | 0.0% | 0.4% |

**Observed and Expected Digit Frequencies**




## Diagnostic Accuracy (Python)

Reference positive: Positive; test positive: Positive. Complete pairs: 36; excluded: 0. One independent observation per row. Predictive values reflect the sample composition.

Confusion Counts

| Classification | Count |
|---|---|
| True positive | 14 |
| False positive | 1 |
| False negative | 2 |
| True negative | 19 |

Sample Accuracy

| Metric | Estimate | CI Lower | CI Upper |
|---|---|---|---|
| Sensitivity | 87.5% | 61.7% | 98.4% |
| Specificity | 95.0% | 75.1% | 99.9% |
| Positive predictive value | 93.3% | 68.1% | 99.8% |
| Negative predictive value | 90.5% | 69.6% | 98.8% |
| Accuracy | 91.7% | 77.5% | 98.2% |

*Note.* 95% exact Clopper-Pearson intervals assume independent observations and a fixed classification rule. PPV, NPV and accuracy describe this sample, not population performance in a case-control sample.

Diagnostic Ratios and Youden Index

| Metric | Estimate | Value or Status |
|---|---|---|
| Positive likelihood ratio | 17.500 | 17.5 |
| Negative likelihood ratio | 0.132 | 0.131579 |
| Diagnostic odds ratio | 133.000 | 133 |
| Youden index | 0.825 | 0.825 |

*Note.* No continuity correction is applied. Infinite means a positive numerator with a zero denominator; undefined includes zero divided by zero or a missing reference margin. Ratio intervals are omitted.

## Chi-Square Post-Tests (Python)

36 complete input rows; 0 excluded; total frequency 106. Frequencies represent independent observations, not sampling weights. Unused and zero-frequency levels are removed. No bootstrap intervals or power calculations are included.

Overall Association

| Total Frequency | Pearson Chi-Square | df | p | Cramers V |
|---|---|---|---|---|
| 106 | 0.925 | 4 | .921 | 0.0660 |

Cell Counts and Adjusted Residual Tests

| Row | Column | Observed | Expected | Adjusted Residual | Cell p | Bonferroni p | Significant |
|---|---|---|---|---|---|---|---|
| A | X | 13.0 | 11.9 | 0.485 | .627 | 1.000 | No |
| A | Y | 10.0 | 12.2 | -0.961 | .337 | 1.000 | No |
| A | Z | 12.0 | 10.9 | 0.492 | .622 | 1.000 | No |
| B | X | 12.0 | 12.6 | -0.244 | .808 | 1.000 | No |
| B | Y | 14.0 | 12.9 | 0.464 | .643 | 1.000 | No |
| B | Z | 11.0 | 11.5 | -0.228 | .819 | 1.000 | No |
| C | X | 11.0 | 11.5 | -0.240 | .810 | 1.000 | No |
| C | Y | 13.0 | 11.9 | 0.494 | .621 | 1.000 | No |
| C | Z | 10.0 | 10.6 | -0.263 | .793 | 1.000 | No |

*Note.* Adjusted standardized residuals account for row and column margins. Cell p-values are two-sided normal approximations, Bonferroni-adjusted across all retained cells independently of pairwise tests.

Pairwise Comparisons

| Compared Dimension | First Level | Second Level | Method | Pearson Chi-Square | df | p | Adjusted p |
|---|---|---|---|---|---|---|---|
| Rows | A | B | Pearson (uncorrected) | 0.69513 | 2 | .706 | 1.000 |
| Rows | A | C | Pearson (uncorrected) | 0.72545 | 2 | .696 | 1.000 |
| Rows | B | C | Pearson (uncorrected) | 0.00138 | 2 | .999 | 1.000 |
| Columns | X | Y | Pearson (uncorrected) | 0.69825 | 2 | .705 | 1.000 |
| Columns | X | Z | Pearson (uncorrected) | 6.64e-4 | 2 | 1.000 | 1.000 |
| Columns | Y | Z | Pearson (uncorrected) | 0.70686 | 2 | .702 | 1.000 |

*Note.* Adjustment: bonferroni; one family across all 6 valid row/column comparisons. Zero margins are removed within each comparison. Two-level dimensions are not retested.

## Decision Curve Analysis (Python)

Compared 2 supplied probability columns on 36 common complete cases (0 rows excluded). Models are not fitted or recalibrated. Thresholds encode the relative consequences of false positives and missed events; the default range is a computational convenience, not a clinical recommendation. Net benefit uses the observed cohort prevalence and assumes independent observations. Bootstrap inference, case-control prevalence correction, model-comparison tests and economic modelling are outside this core.

Common Complete-Case Cohort

| Included | Excluded | Reference Positive | Sample Prevalence |
|---|---|---|---|
| 36 | 0 | 16 | 44.4% |

Threshold Results

| Strategy | Threshold | TP | FP | Net Benefit | Interventions Avoided per 100 |
|---|---|---|---|---|---|
| Model: RiskA | 0.0500 | 16 | 20 | 0.41520 | 0.00 |
| Model: RiskA | 0.0600 | 16 | 20 | 0.40898 | 0.00 |
| Model: RiskA | 0.0700 | 16 | 20 | 0.40263 | 0.00 |
| Model: RiskA | 0.0800 | 16 | 19 | 0.39855 | 2.78 |
| Model: RiskA | 0.0900 | 16 | 16 | 0.40049 | 11.11 |
| Model: RiskA | 0.1000 | 16 | 15 | 0.39815 | 13.89 |
| Model: RiskA | 0.1100 | 16 | 14 | 0.39638 | 16.67 |
| Model: RiskA | 0.1200 | 16 | 13 | 0.39520 | 19.44 |
| Model: RiskA | 0.1300 | 16 | 12 | 0.39464 | 22.22 |
| Model: RiskA | 0.1400 | 16 | 12 | 0.39018 | 22.22 |
| Model: RiskA | 0.1500 | 16 | 11 | 0.39052 | 25.00 |
| Model: RiskA | 0.1600 | 16 | 11 | 0.38624 | 25.00 |
| Model: RiskA | 0.1700 | 16 | 10 | 0.38755 | 27.78 |
| Model: RiskA | 0.1800 | 16 | 10 | 0.38347 | 27.78 |
| Model: RiskA | 0.1900 | 16 | 10 | 0.37929 | 27.78 |
| Model: RiskA | 0.2000 | 16 | 10 | 0.37500 | 27.78 |
| Model: RiskA | 0.2100 | 16 | 10 | 0.37060 | 27.78 |
| Model: RiskA | 0.2200 | 16 | 10 | 0.36610 | 27.78 |
| Model: RiskA | 0.2300 | 16 | 10 | 0.36147 | 27.78 |
| Model: RiskA | 0.2400 | 16 | 10 | 0.35673 | 27.78 |
| Model: RiskA | 0.2500 | 16 | 10 | 0.35185 | 27.78 |
| Model: RiskA | 0.2600 | 16 | 10 | 0.34685 | 27.78 |
| Model: RiskA | 0.2700 | 16 | 9 | 0.35198 | 30.56 |
| Model: RiskA | 0.2800 | 16 | 9 | 0.34722 | 30.56 |
| Model: RiskA | 0.2900 | 16 | 9 | 0.34233 | 30.56 |
| Model: RiskA | 0.3000 | 16 | 8 | 0.34921 | 33.33 |
| Model: RiskA | 0.3100 | 16 | 8 | 0.34461 | 33.33 |
| Model: RiskA | 0.3200 | 16 | 8 | 0.33987 | 33.33 |
| Model: RiskA | 0.3300 | 16 | 6 | 0.36235 | 38.89 |
| Model: RiskA | 0.3400 | 16 | 6 | 0.35859 | 38.89 |
| Model: RiskA | 0.3500 | 16 | 6 | 0.35470 | 38.89 |
| Model: RiskA | 0.3600 | 16 | 5 | 0.36632 | 41.67 |
| Model: RiskA | 0.3700 | 16 | 4 | 0.37919 | 44.44 |
| Model: RiskA | 0.3800 | 16 | 3 | 0.39337 | 47.22 |
| Model: RiskA | 0.3900 | 16 | 3 | 0.39117 | 47.22 |
| Model: RiskA | 0.4000 | 16 | 2 | 0.40741 | 50.00 |
| Model: RiskA | 0.4100 | 15 | 2 | 0.37806 | 46.00 |
| Model: RiskA | 0.4200 | 15 | 2 | 0.37644 | 46.16 |
| Model: RiskA | 0.4300 | 15 | 2 | 0.37476 | 46.32 |
| Model: RiskA | 0.4400 | 15 | 2 | 0.37302 | 46.46 |
| Model: RiskA | 0.4500 | 15 | 2 | 0.37121 | 46.60 |
| Model: RiskA | 0.4600 | 15 | 2 | 0.36934 | 46.74 |

Threshold Results

| Strategy | Threshold | TP | FP | Net Benefit | Interventions Avoided per 100 |
|---|---|---|---|---|---|
| Model: RiskA | 0.4700 | 15 | 2 | 0.36740 | 46.87 |
| Model: RiskA | 0.4800 | 15 | 2 | 0.36538 | 46.99 |
| Model: RiskA | 0.4900 | 14 | 1 | 0.36220 | 47.00 |
| Model: RiskA | 0.5000 | 14 | 1 | 0.36111 | 47.22 |
| Model: RiskB | 0.0500 | 16 | 15 | 0.42251 | 13.89 |
| Model: RiskB | 0.0600 | 16 | 15 | 0.41785 | 13.89 |
| Model: RiskB | 0.0700 | 16 | 15 | 0.41308 | 13.89 |
| Model: RiskB | 0.0800 | 16 | 15 | 0.40821 | 13.89 |
| Model: RiskB | 0.0900 | 16 | 15 | 0.40324 | 13.89 |
| Model: RiskB | 0.1000 | 16 | 15 | 0.39815 | 13.89 |
| Model: RiskB | 0.1100 | 16 | 15 | 0.39295 | 13.89 |
| Model: RiskB | 0.1200 | 16 | 14 | 0.39141 | 16.67 |
| Model: RiskB | 0.1300 | 16 | 14 | 0.38633 | 16.67 |
| Model: RiskB | 0.1400 | 16 | 13 | 0.38566 | 19.44 |
| Model: RiskB | 0.1500 | 16 | 12 | 0.38562 | 22.22 |
| Model: RiskB | 0.1600 | 16 | 11 | 0.38624 | 25.00 |
| Model: RiskB | 0.1700 | 16 | 11 | 0.38186 | 25.00 |
| Model: RiskB | 0.1800 | 16 | 10 | 0.38347 | 27.78 |
| Model: RiskB | 0.1900 | 16 | 10 | 0.37929 | 27.78 |
| Model: RiskB | 0.2000 | 16 | 10 | 0.37500 | 27.78 |
| Model: RiskB | 0.2100 | 16 | 9 | 0.37799 | 30.56 |
| Model: RiskB | 0.2200 | 16 | 9 | 0.37393 | 30.56 |
| Model: RiskB | 0.2300 | 16 | 9 | 0.36977 | 30.56 |
| Model: RiskB | 0.2400 | 16 | 8 | 0.37427 | 33.33 |
| Model: RiskB | 0.2500 | 16 | 8 | 0.37037 | 33.33 |
| Model: RiskB | 0.2600 | 16 | 8 | 0.36637 | 33.33 |
| Model: RiskB | 0.2700 | 16 | 7 | 0.37253 | 36.11 |
| Model: RiskB | 0.2800 | 16 | 7 | 0.36883 | 36.11 |
| Model: RiskB | 0.2900 | 16 | 7 | 0.36502 | 36.11 |
| Model: RiskB | 0.3000 | 16 | 7 | 0.36111 | 36.11 |
| Model: RiskB | 0.3100 | 16 | 7 | 0.35709 | 36.11 |
| Model: RiskB | 0.3200 | 16 | 7 | 0.35294 | 36.11 |
| Model: RiskB | 0.3300 | 16 | 7 | 0.34867 | 36.11 |
| Model: RiskB | 0.3400 | 16 | 6 | 0.35859 | 38.89 |
| Model: RiskB | 0.3500 | 16 | 5 | 0.36966 | 41.67 |
| Model: RiskB | 0.3600 | 16 | 5 | 0.36632 | 41.67 |
| Model: RiskB | 0.3700 | 16 | 3 | 0.39550 | 47.22 |
| Model: RiskB | 0.3800 | 16 | 3 | 0.39337 | 47.22 |
| Model: RiskB | 0.3900 | 16 | 1 | 0.42668 | 52.78 |
| Model: RiskB | 0.4000 | 16 | 1 | 0.42593 | 52.78 |
| Model: RiskB | 0.4100 | 15 | 1 | 0.39736 | 48.78 |
| Model: RiskB | 0.4200 | 15 | 1 | 0.39655 | 48.94 |

Threshold Results

| Strategy | Threshold | TP | FP | Net Benefit | Interventions Avoided per 100 |
|---|---|---|---|---|---|
| Model: RiskB | 0.4300 | 15 | 1 | 0.39571 | 49.10 |
| Model: RiskB | 0.4400 | 15 | 1 | 0.39484 | 49.24 |
| Model: RiskB | 0.4500 | 15 | 1 | 0.39394 | 49.38 |
| Model: RiskB | 0.4600 | 15 | 1 | 0.39300 | 49.52 |
| Model: RiskB | 0.4700 | 15 | 1 | 0.39203 | 49.65 |
| Model: RiskB | 0.4800 | 15 | 1 | 0.39103 | 49.77 |
| Model: RiskB | 0.4900 | 15 | 1 | 0.38998 | 49.89 |
| Model: RiskB | 0.5000 | 15 | 1 | 0.38889 | 50.00 |
| Reference: treat all | 0.0500 | 16 | 20 | 0.41520 | 0.00 |
| Reference: treat all | 0.0600 | 16 | 20 | 0.40898 | 0.00 |
| Reference: treat all | 0.0700 | 16 | 20 | 0.40263 | 0.00 |
| Reference: treat all | 0.0800 | 16 | 20 | 0.39614 | 0.00 |
| Reference: treat all | 0.0900 | 16 | 20 | 0.38950 | 0.00 |
| Reference: treat all | 0.1000 | 16 | 20 | 0.38272 | 0.00 |
| Reference: treat all | 0.1100 | 16 | 20 | 0.37578 | 0.00 |
| Reference: treat all | 0.1200 | 16 | 20 | 0.36869 | 0.00 |
| Reference: treat all | 0.1300 | 16 | 20 | 0.36143 | 0.00 |
| Reference: treat all | 0.1400 | 16 | 20 | 0.35401 | 0.00 |
| Reference: treat all | 0.1500 | 16 | 20 | 0.34641 | 0.00 |
| Reference: treat all | 0.1600 | 16 | 20 | 0.33862 | 0.00 |
| Reference: treat all | 0.1700 | 16 | 20 | 0.33066 | 0.00 |
| Reference: treat all | 0.1800 | 16 | 20 | 0.32249 | 0.00 |
| Reference: treat all | 0.1900 | 16 | 20 | 0.31413 | 0.00 |
| Reference: treat all | 0.2000 | 16 | 20 | 0.30556 | 0.00 |
| Reference: treat all | 0.2100 | 16 | 20 | 0.29677 | 0.00 |
| Reference: treat all | 0.2200 | 16 | 20 | 0.28775 | 0.00 |
| Reference: treat all | 0.2300 | 16 | 20 | 0.27850 | 0.00 |
| Reference: treat all | 0.2400 | 16 | 20 | 0.26901 | 0.00 |
| Reference: treat all | 0.2500 | 16 | 20 | 0.25926 | 0.00 |
| Reference: treat all | 0.2600 | 16 | 20 | 0.24925 | 0.00 |
| Reference: treat all | 0.2700 | 16 | 20 | 0.23896 | 0.00 |
| Reference: treat all | 0.2800 | 16 | 20 | 0.22840 | 0.00 |
| Reference: treat all | 0.2900 | 16 | 20 | 0.21753 | 0.00 |
| Reference: treat all | 0.3000 | 16 | 20 | 0.20635 | 0.00 |
| Reference: treat all | 0.3100 | 16 | 20 | 0.19485 | 0.00 |
| Reference: treat all | 0.3200 | 16 | 20 | 0.18301 | 0.00 |
| Reference: treat all | 0.3300 | 16 | 20 | 0.17081 | 0.00 |
| Reference: treat all | 0.3400 | 16 | 20 | 0.15825 | 0.00 |
| Reference: treat all | 0.3500 | 16 | 20 | 0.14530 | 0.00 |
| Reference: treat all | 0.3600 | 16 | 20 | 0.13194 | 0.00 |
| Reference: treat all | 0.3700 | 16 | 20 | 0.11817 | 0.00 |
| Reference: treat all | 0.3800 | 16 | 20 | 0.10394 | 0.00 |

Threshold Results

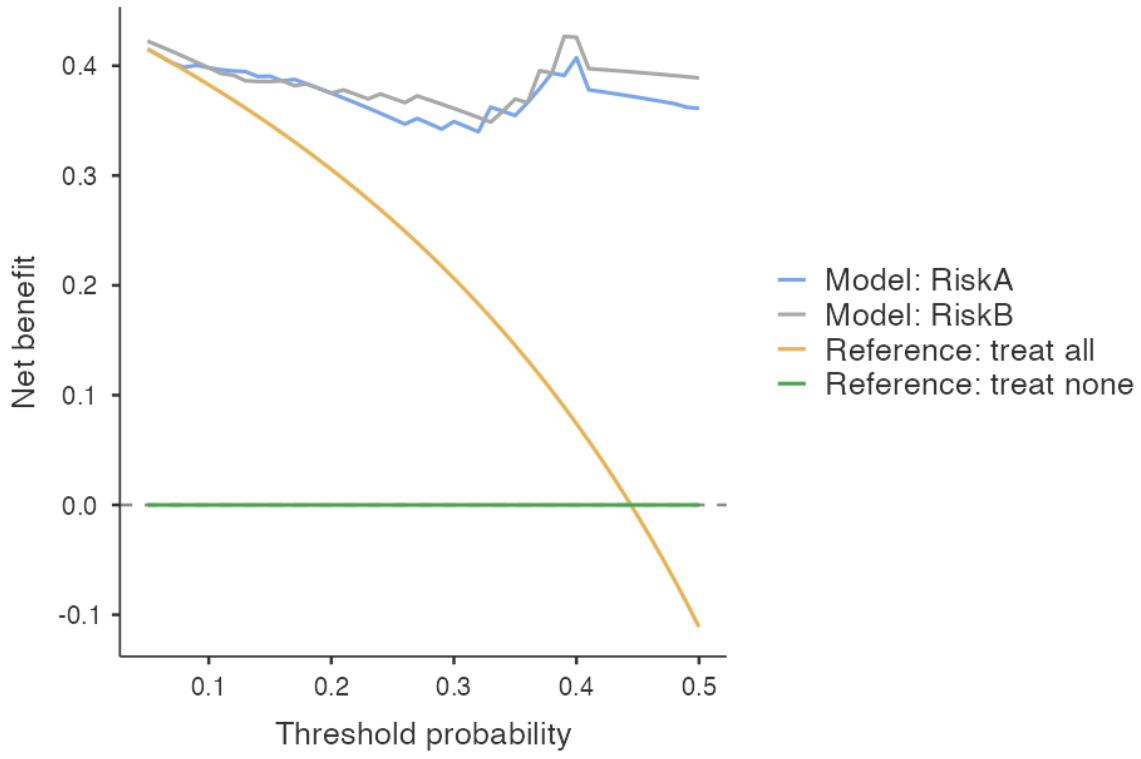
| Strategy | Threshold | TP | FP | Net Benefit | Interventions Avoided per 100 |
|---|---|---|---|---|---|
| Reference: treat all | 0.3900 | 16 | 20 | 0.08925 | 0.00 |
| Reference: treat all | 0.4000 | 16 | 20 | 0.07407 | 0.00 |
| Reference: treat all | 0.4100 | 16 | 20 | 0.05838 | 0.00 |
| Reference: treat all | 0.4200 | 16 | 20 | 0.04215 | 0.00 |
| Reference: treat all | 0.4300 | 16 | 20 | 0.02534 | 0.00 |
| Reference: treat all | 0.4400 | 16 | 20 | 0.00794 | 0.00 |
| Reference: treat all | 0.4500 | 16 | 20 | -0.01010 | 0.00 |
| Reference: treat all | 0.4600 | 16 | 20 | -0.02881 | 0.00 |
| Reference: treat all | 0.4700 | 16 | 20 | -0.04822 | 0.00 |
| Reference: treat all | 0.4800 | 16 | 20 | -0.06838 | 0.00 |
| Reference: treat all | 0.4900 | 16 | 20 | -0.08932 | 0.00 |
| Reference: treat all | 0.5000 | 16 | 20 | -0.11111 | 0.00 |
| Reference: treat none | 0.0500 | 0 | 0 | 0.00000 | -788.89 |
| Reference: treat none | 0.0600 | 0 | 0 | 0.00000 | -640.74 |
| Reference: treat none | 0.0700 | 0 | 0 | 0.00000 | -534.92 |
| Reference: treat none | 0.0800 | 0 | 0 | 0.00000 | -455.56 |
| Reference: treat none | 0.0900 | 0 | 0 | 0.00000 | -393.83 |
| Reference: treat none | 0.1000 | 0 | 0 | 0.00000 | -344.44 |
| Reference: treat none | 0.1100 | 0 | 0 | 0.00000 | -304.04 |
| Reference: treat none | 0.1200 | 0 | 0 | 0.00000 | -270.37 |
| Reference: treat none | 0.1300 | 0 | 0 | 0.00000 | -241.88 |
| Reference: treat none | 0.1400 | 0 | 0 | 0.00000 | -217.46 |
| Reference: treat none | 0.1500 | 0 | 0 | 0.00000 | -196.30 |
| Reference: treat none | 0.1600 | 0 | 0 | 0.00000 | -177.78 |
| Reference: treat none | 0.1700 | 0 | 0 | 0.00000 | -161.44 |
| Reference: treat none | 0.1800 | 0 | 0 | 0.00000 | -146.91 |
| Reference: treat none | 0.1900 | 0 | 0 | 0.00000 | -133.92 |
| Reference: treat none | 0.2000 | 0 | 0 | 0.00000 | -122.22 |
| Reference: treat none | 0.2100 | 0 | 0 | 0.00000 | -111.64 |
| Reference: treat none | 0.2200 | 0 | 0 | 0.00000 | -102.02 |
| Reference: treat none | 0.2300 | 0 | 0 | 0.00000 | -93.24 |
| Reference: treat none | 0.2400 | 0 | 0 | 0.00000 | -85.19 |
| Reference: treat none | 0.2500 | 0 | 0 | 0.00000 | -77.78 |
| Reference: treat none | 0.2600 | 0 | 0 | 0.00000 | -70.94 |
| Reference: treat none | 0.2700 | 0 | 0 | 0.00000 | -64.61 |
| Reference: treat none | 0.2800 | 0 | 0 | 0.00000 | -58.73 |
| Reference: treat none | 0.2900 | 0 | 0 | 0.00000 | -53.26 |
| Reference: treat none | 0.3000 | 0 | 0 | 0.00000 | -48.15 |
| Reference: treat none | 0.3100 | 0 | 0 | 0.00000 | -43.37 |
| Reference: treat none | 0.3200 | 0 | 0 | 0.00000 | -38.89 |
| Reference: treat none | 0.3300 | 0 | 0 | 0.00000 | -34.68 |
| Reference: treat none | 0.3400 | 0 | 0 | 0.00000 | -30.72 |

Threshold Results

| Strategy | Threshold | TP | FP | Net Benefit | Interventions Avoided per 100 |
|---|---|---|---|---|---|
| Reference: treat none | 0.3500 | 0 | 0 | 0.00000 | -26.98 |
| Reference: treat none | 0.3600 | 0 | 0 | 0.00000 | -23.46 |
| Reference: treat none | 0.3700 | 0 | 0 | 0.00000 | -20.12 |
| Reference: treat none | 0.3800 | 0 | 0 | 0.00000 | -16.96 |
| Reference: treat none | 0.3900 | 0 | 0 | 0.00000 | -13.96 |
| Reference: treat none | 0.4000 | 0 | 0 | 0.00000 | -11.11 |
| Reference: treat none | 0.4100 | 0 | 0 | 0.00000 | -8.40 |
| Reference: treat none | 0.4200 | 0 | 0 | 0.00000 | -5.82 |
| Reference: treat none | 0.4300 | 0 | 0 | 0.00000 | -3.36 |
| Reference: treat none | 0.4400 | 0 | 0 | 0.00000 | -1.01 |
| Reference: treat none | 0.4500 | 0 | 0 | 0.00000 | 1.23 |
| Reference: treat none | 0.4600 | 0 | 0 | 0.00000 | 3.38 |
| Reference: treat none | 0.4700 | 0 | 0 | 0.00000 | 5.44 |
| Reference: treat none | 0.4800 | 0 | 0 | 0.00000 | 7.41 |
| Reference: treat none | 0.4900 | 0 | 0 | 0.00000 | 9.30 |
| Reference: treat none | 0.5000 | 0 | 0 | 0.00000 | 11.11 |

*Note.* Positive classification is predicted probability >= threshold. Interventions avoided per 100 are relative to treat all at the same threshold; negative values indicate more interventions at equivalent net benefit. All strategies use the same complete-case cohort. No uncertainty intervals are included.

**Net Benefit Curves**




# References


[1] R Core Team (2026). *R: A Language and environment for statistical computing*. (Version 4.6) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2026-05-11).

[2] The jamovi project (2026). *jamovi*. (Version 28) [Computer Software]. Retrieved from https://www.jamovi.org.